The bottom of the spectrum of the critical GCD matrix

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Abstract. For $N \geq 1$ let K_N be the $N \times N$ matrix with entries $K(m,n) = \gcd(m,n)/\sqrt{mn}$, $1 \leq m,n \leq N$. The matrix is positive definite for every N, and its quadratic form is the Gál-type GCD sum at the critical exponent $\alpha = 1/2$; while the top of the spectrum is governed by a well-developed theory, the decay of the smallest eigenvalue $\lambda_{\min}(N)$ appears not to have been studied. We bound it on both sides, unconditionally. From above: whenever the primorial $\prod_{p \leq y} p$ is at most N, the Möbius-signed indicator of its divisors achieves Rayleigh quotient exactly $\prod_{p \leq y}\left(1 - p^{-1/2}\right)$, whence $\lambda_{\min}(N) \leq \exp\left(-(2+o(1))\sqrt{\log N}/\log\log N\right)$; the quadratic-form identity behind the bound is machine-checked in Lean 4, and no test family of product (tensor) type can beat $\exp\left(-(4+o(1))\sqrt{\log N}/\log\log N\right)$. From below: the explicit Möbius inverse hides no cancellation. It factors as $K_N^{-1} = \Delta_\lambda\,|K_N^{-1}|\,\Delta_\lambda$ with Δ_λ the diagonal matrix of Liouville signs, so the minimizing eigenvector carries the exact Liouville sign pattern for every N (Perron–Frobenius), and a weighted Schur test on the nonnegative factor, with per-prime geometric weights scheduled in proportion to the logarithmic budget $\sum_{p|n} \log p \leq \log n$, gives $\lambda_{\min}(N) \geq \exp\left(-(2+o(1))\sqrt{\log N/\log\log N}\right)$. Together the two bounds settle the exponent, $-\log \lambda_{\min}(N) = (\log N)^{1/2+o(1)}$, confine the slowly varying factor to a corridor of relative width $\sqrt{\log\log N}$, and exclude the pure law $\exp\left(-(c+o(1))\sqrt{\log N}\right)$: the exponential-rate constants fitted at finite N are transient. On the computational side, the inverse of the Cholesky-type Gram factor of K_N is an explicit sparse Möbius matrix, which turns $1/\lambda_{\min}$ into a dominant eigenvalue accessible to Lanczos iteration; we compute $\lambda_{\min}(N)$ with certified residuals up to $N = 1.3 \times 10^7$. The data reject the law $c(\log N)^{-2}$, reject the product-family rate, and land inside the proved corridor. Finally, an exact symmetry is proved (also machine-checked): conjugation by the diagonal matrix of Liouville signs realizes the half-turn of every prime circle underlying the kernel. It is set beside an unexplained empirical regularity: $\lambda_{\min} \cdot \lambda_{\max} \approx 0.389$, constant to $\pm 0.1\%$ across the last two decades of the computed range.

1 Introduction

1.1 The object

For an integer $N \geq 1$ consider the $N \times N$ matrix

$$K_N(m,n) = \frac{\gcd(m,n)}{\sqrt{mn}}, \quad 1 \leq m,n \leq N.$$

Since $\gcd(m,n) = \sum_{d|m,\, d|n} \varphi(d)$, the matrix admits the Gram factorization

$$K_N = C_N C_N^{\top}, \quad C_N = D\,B\,\mathrm{diag}\left(\sqrt{\varphi(1)}, \dots, \sqrt{\varphi(N)}\right),$$

where $D = \mathrm{diag}(1/\sqrt{n})$ and B is the 0–1 divisibility matrix $B[n, d] = 1 \Leftrightarrow d \mid n$ (unit lower triangular in the natural order). In particular K_N is positive semidefinite, and Smith's determinant evaluation [Smi76]

$$\det K_N = \prod_{n \le N} \frac{\varphi(n)}{n} \neq 0$$

upgrades this to positive definiteness. Write $\lambda_{\min}(N) \le \ldots \le \lambda_{\max}(N)$ for the spectrum. This paper is about the bottom edge:

At what rate does $\lambda_{\min}(N) \to 0$, and by what mechanism?

1.2 Why the exponent $1/2$ is critical

The quadratic form of K_N is the GCD sum $\sum_{m,n} \gcd(m, n)^{2\alpha}/(mn)^{\alpha}$ at $\alpha = 1/2$, the critical exponent of the Gál-sum literature. For $\alpha = 1$, Gál's classical theorem [Gal49] gives the optimal order $(\log \log N)^2$ for the normalized sum over arbitrary sets of N integers. At $\alpha = 1/2$ the problem changes character because $\sum_p p^{-2\alpha}$ diverges exactly there: Dyer and Harman [DH86] obtained $\exp(c \log N / \log \log N)$; Aistleitner, Berkes and Seip [ABS15] solved the Dyer–Harman problem with a bound of the shape $\exp(C\sqrt{\log N \log \log \log N})$, noting that nothing below $\exp\left(2\sqrt{\log N / \log \log N}\right)$ is possible; Bondarenko and Seip [BS15] sharpened the upper bound to the shape $\exp\left(C\sqrt{\log N \log \log \log N / \log \log N}\right)$, later fixing the optimal constant with consequences for large values of ζ on the critical line [BS17]. All of this concerns the **top** of the spectrum over extremal sets; the interval $\{1, \ldots, N\}$ is far from extremal (its top eigenvalue grows only polylogarithmically on the computed range; see Section 5).

Two more boundaries meet at the same exponent. Lindqvist and Seip [LS98] proved positive definiteness and sharp eigenvalue bounds for the family $\left[\gcd(m, n)^{2\alpha}/(mn)^{\alpha}\right]$ for $\alpha > 1/2$, stopping exactly at the critical case. And since $\gcd(m, n)/\sqrt{mn} = \sqrt{mn}/[m, n]$, our kernel is the boundary case $E(1/2, 1)$ of the family $E(\sigma, \tau) = (n^{\sigma} m^{\sigma}/[n, m]^{\tau})$ studied by Hilberdink and Pushnitski [HP22] (see also arXiv:2401.06892): for $\rho = \tau - 2\sigma > 0$ the operator on $\ell^2(\mathbb{N})$ is compact with power-law eigenvalue asymptotics, and the hypotheses exclude precisely $\rho = 0$. On that boundary the operator is unbounded above and 0 is a spectral accumulation point from below: the truncations K_N straddle a non-compact critical operator, the two spectral edges open up as $N \to \infty$, and quantitative rates become the question. For the bottom edge on the interval at $\alpha = 1/2$, we have found no source stating even a conjectural rate; to the author's knowledge the question appears here for the first time (see Section 7 for the full sweep, including the smallest-eigenvalue literature for unnormalized GCD matrices [HL04]).

1.3 Results

The main theorem is an explicit, unconditional upper bound.

Theorem A (primorial bound; Section 2). *Whenever $\prod_{p \le y} p \le N$, the Möbius-signed indicator of the divisors of the primorial achieves Rayleigh quotient exactly $\prod_{p \le y}\left(1 - p^{-1/2}\right)$ for K_N. Consequently*

$$\lambda_{\min}(N) \le \exp\left(-(2 + o(1))\frac{\sqrt{\log N}}{\log \log N}\right).$$

The Rayleigh-quotient identity is exact, and is machine-checked in Lean 4 (Section 2.5). The asymptotic evaluation uses only Mertens' theorem and the prime number theorem; a fully explicit variant (constant 0.99, valid for $N \geq 3.3 \times 10^7$) follows from Rosser–Schoenfeld bounds. Already this bound decays faster than every power of $\log N$.

A companion barrier shows the construction is essentially the best of its kind: no test vector of product (tensor) type over the primes can achieve a Rayleigh quotient below $\exp\left(-(4+o(1))\sqrt{\log N}/\log\log N\right)$ (Section 3).

The matching lower bound comes from the inverse. Its entries are explicit, and they hide no cancellation:

Theorem B (Liouville gauge; Section 4). *The inverse factors as* $K_N^{-1} = \Delta_\lambda\,|K_N^{-1}|\,\Delta_\lambda$*, where* $\Delta_\lambda = \mathrm{diag}(\lambda(1), ..., \lambda(N))$ *is the diagonal matrix of Liouville signs and* $|K_N^{-1}|$ *is entrywise nonnegative and irreducible. Consequently* $\lambda_{\min}(N)$ *is a simple eigenvalue for every* N*, and its eigenvector has sign pattern exactly* $\lambda(n)$*.*

Theorem C (lower bound; Section 4). *Unconditionally,*

$$\lambda_{\min}(N) \geq \exp\left(-(2+o(1))\sqrt{\log\frac{N}{\log}\log N}\right).$$

The weight is a per-prime scheduled Schur test on $|K_N^{-1}|$; the inputs are Mertens, Rosser–Schoenfeld, and Chebyshev-grade bounds only — neither the prime number theorem nor Robin's bound is used.

Theorems A and C together settle the exponent and exclude a law:

Corollary D (the rate; Section 4). $-\log\lambda_{\min}(N) = (\log N)^{1/2+o(1)}$, *and* $\lambda_{\min}(N) = \exp\left(-(c+o(1))\sqrt{\log N}\right)$ *holds for no constant* $c > 0$*.*

What remains open is the slowly varying factor: the corridor $-\log\lambda_{\min} \in \left[0.99\sqrt{\log N}/\log\log N, (2+o(1))\sqrt{\log N/\log\log N}\right]$ has relative width of shape $\sqrt{\log\log N}$, and the numerics of Section 5 (exact sparse computations of $\lambda_{\min}$ to $N = 1.3 \times 10^7$, which reject both the two-parameter law $c(\log N)^{-2}$ and the product-family rate on the computed window) measure the window-effective exponential constant at ≈ 1.36–1.41, inside the corridor; by Corollary D its measured decline is forced.

Finally, Section 6 records an exact symmetry and an unexplained regularity. The half-turn $\theta_p \mapsto \theta_p + \pi$ of every circle in the prime-torus (polydisc) representation of the kernel acts on K_N as conjugation by the diagonal matrix of Liouville signs $\mathrm{diag}(\lambda(1), ..., \lambda(N))$: an involutive similarity, also machine-checked, carrying the same signs Theorem B proves are the minimizer's. Beside it sits the most robust unexplained pattern in the data: $\lambda_{\min}(N) \cdot \lambda_{\max}(N) \approx 0.389$, constant to $\pm 0.1\%$ from $N \approx 2 \times 10^5$ to 1.3×10^7 while each factor moves by a factor of about 1.9.

1.4 Methodology and claim status

The order of discovery is worth recording because it shaped the paper. Theorem A was **conjectured from computed eigenvector portraits**: dense spectra up to $N = 25\,600$ show the minimizing eigenvector concentrating on small, highly divisible integers, with signs following the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ at mass-weighted agreement above 0.999

(Figure 1, right panel). The primorial test vector is the simplest exact structure matching that portrait. The identity was then derived on paper, validated numerically at machine precision (worst relative deviation 4.2×10^{-15} across four independent evaluation routes on a 19-point grid), and finally formalized in Lean 4 against a pinned Mathlib. The lower side repeated the pattern: the Liouville sign structure of Theorem B was first a measured fact (mass-weighted sign agreement above 0.999; the identity $\lambda_{\max}(|K_N^{-1}|) = \lambda_{\max}(K_N^{-1})$ holding to all printed digits) and became a three-line gauge identity once the entry formula of the inverse was written down. Correctness of the machine-checked claims does not rest on trusting any author, human or artificial: the proofs are checked by the Lean kernel, the repository [RV26] enforces a zero-`sorry` policy in continuous integration, and the named theorems carry axiom audits.

Throughout the paper every substantive claim carries one of four labels: **machine-checked** (proved in Lean 4 in [RV26]; exact declaration names are cited), **classical** (standard literature, cited), **derived** (complete paper proof given here, not formalized), and **numerical** (reproducible computation; no claim beyond the computed range).

2 The primorial bound

Throughout this section S is a finite set of primes and $Q = \prod_{p \in S} p$ its primorial-type product. For $T \subseteq S$ write $q_T = \prod_{p \in T} p$; the $2^{|S|}$ integers q_T are exactly the divisors of Q, and they are pairwise distinct by unique factorization.

2.1 The per-prime factorization on squarefree support

Lemma 1 (per-prime factorization; machine-checked). *Let S be a finite set of primes and $T, T' \subseteq S$. Then*

$$K(q_T, q_{T'}) = \prod_{p \in S} e_p, \quad e_p = \begin{cases} 1 & \text{if } p \in T \Leftrightarrow p \in T' \\ p^{-1/2} & \text{otherwise.} \end{cases}$$

Proof. By coprimality of distinct primes, $\gcd(q_T, q_{T'}) = q_{T \cap T'}$, while $\sqrt{q_T\, q_{T'}} = q_{T \cap T'} \cdot \prod_{p \in T \Delta T'} \sqrt{p}$, where $T \Delta T'$ is the symmetric difference. Dividing, each prime in $T \Delta T'$ contributes exactly $p^{-1/2}$ and every other prime of S contributes 1. □

In other words, the restriction of K_N to the divisors of Q is the $|S|$-fold tensor product

$$K_N \mid_{\operatorname{div} Q} = \bigotimes_{p \in S} T_2(p^{-1/2}), \quad T_2(a) = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix},$$

the 2×2 instance of the Kac–Murdock–Szegő structure that the kernel carries on every smooth (friable) support. Formally this is `gcdKernelEntry_prod_prod` in [RV26].

2.2 The signed double sum

Lemma 2 (signed powerset double sum; machine-checked). *For any weights $g : \mathbb{P} \to \mathbb{R}$ and any finite prime set S,*

$$\sum_{T \subseteq S} \sum_{T' \subseteq S} (-1)^{|T|}(-1)^{|T'|} \prod_{p \in S} e_p(T, T') = \prod_{p \in S} (2 - 2g(p)),$$

where $e_p(T, T') = 1$ if $p \in T \Leftrightarrow p \in T'$ and $e_p(T, T') = g(p)$ otherwise.

Proof. Induction on $|S|$. For $S = \emptyset$ both sides equal 1. Suppose the identity holds for S_0 and let $q \notin S_0$ be a further prime. Split each of $T, T' \subseteq S_0 \cup \{q\}$ according to whether it contains q. In the four resulting blocks the factor at q is 1 (q in neither), $g(q)$ (q in exactly one, twice), and 1 (q in both), while the signs contribute $+1, -1, -1, +1$ respectively and the factors at the primes of S_0 are unchanged. Hence the double sum over $S_0 \cup \{q\}$ equals $(1 - g(q) - g(q) + 1) = (2 - 2g(q))$ times the double sum over S_0, and the induction closes. □

This is pure sign combinatorics, with no arithmetic input at all; it is `signed_powerset_double_sum` in [RV26].

2.3 The bound

Theorem 1 (primorial Rayleigh bound; machine-checked). *Let S be a finite set of primes with $\prod_{p \in S} p \leq N$. Then*

$$\lambda_{\min}(K_N) \leq \prod_{p \in S} \big(1 - p^{-1/2}\big),$$

and the bound is attained exactly as the Rayleigh quotient of an explicit vector: the Möbius-signed indicator of the divisors of the primorial.

Proof. Every divisor $q_T \leq Q \leq N$, so the divisors of Q embed into $\{1, ..., N\}$. Define $x \in \mathbb{R}^N$ by $x_{q_T} = (-1)^{|T|} = \mu(q_T)$ for $T \subseteq S$ and $x_n = 0$ off the divisors of Q; then $x \neq 0$ (indeed $x_1 = 1$) and $x^\top x = 2^{|S|}$. By Lemma 1 and Lemma 2 with $g(p) = p^{-1/2}$,

$$x^\top K_N x = \sum_{T, T' \subseteq S} (-1)^{|T| + |T'|} K(q_T, q_{T'}) = \prod_{p \in S} \big(2 - 2p^{-1/2}\big).$$

Since $\prod_{p \in S} \big(2 - 2p^{-1/2}\big) = 2^{|S|} \prod_{p \in S} \big(1 - p^{-1/2}\big)$, the Rayleigh quotient of x equals $\prod_{p \in S} \big(1 - p^{-1/2}\big)$ exactly, and $\lambda_{\min} \leq R(x)$. □

Remark. In the tensor reading, $x = \bigotimes_p (1, -1)$, and per prime the quotient is $((1, -1) T_2(a) (1, -1)^\top)/2 = (2 - 2a)/2 = 1 - a$, the smallest eigenvalue of $T_2(a)$. So the witness is the exact bottom eigenvector of the **restricted** kernel; the inequality in Theorem 1 gives away only the interaction between the primorial divisors and the rest of $\{1, ..., N\}$. The support is Gál's classical extremal set [Gal49] for GCD sums; the literature uses it with positive signs for the **top** of the spectrum. The Möbius signing, and the bottom-edge application, appear to be new.

2.4 The asymptotic evaluation

Corollary 1 (derived; classical inputs as marked). *As $N \to \infty$,*

$$\lambda_{\min}(K_N) \le \exp\left(-(2+o(1))\frac{\sqrt{\log N}}{\log\log N}\right).$$

Proof. Let $y = y(N)$ be the largest prime with $\theta(y) = \sum_{p \le y} \log p \le \log N$, and take $S = \{p \le y\}$ in Theorem 1, so that the primorial constraint holds by construction. By the prime number theorem in the form $\theta(y) \sim y$ [classical], $y = (1+o(1))\log N$. Taking logarithms in Theorem 1,

$$-\log \lambda_{\min}(K_N) \ge \sum_{p \le y} -\log\bigl(1 - p^{-1/2}\bigr) = \sum_{p \le y} p^{-1/2} + O\left(\sum_{p \le y} \frac{1}{p}\right),$$

using $-\log(1-t) = t + O\bigl(t^2\bigr)$ for $0 < t \le 2^{-1/2}$. The error term is $O(\log\log y)$ by Mertens' second theorem [classical; see Ten15, Ch. I.1]. For the main term, partial summation with $\pi(t) = (1+o(1))t/\log t$ [classical, PNT] gives

$$\sum_{p \le y} p^{-1/2} = \frac{\pi(y)}{\sqrt{y}} + \frac{1}{2}\int_2^y \pi(t)\, t^{-3/2}\,\mathrm{d}t = (2+o(1))\frac{\sqrt{y}}{\log y},$$

since both $\pi(y)/\sqrt{y}$ and $\frac{1}{2}\int_2^y t^{-1/2}(\log t)^{-1}\,\mathrm{d}t$ are $(1+o(1))\sqrt{y}/\log y$. Substituting $y = (1+o(1))\log N$ yields $\sqrt{y}/\log y = (1+o(1))\sqrt{\log N}/\log\log N$, which dominates the $O(\log\log y) = O(\log\log\log N)$ error, and the claim follows. □

Remark. Explicit version [derived]. No PNT-strength input is actually required for a numerically explicit bound: using only Rosser–Schoenfeld bounds [RS62] ($\theta(y) \le 1.01624\,y$ for all y, and $\pi(y) \ge y/\log y$ for $y \ge 17$), the same test vector gives

$$\lambda_{\min}(K_N) \le \exp\left(-0.99\,\frac{\sqrt{\log N}}{\log\log N}\right) \quad \text{for all } N \ge 3.3 \times 10^7.$$

In particular Corollary 1 settles, on the upper side, that all laws of the form $c(\log N)^{-A}$ are untenable for $\lambda_{\min}$, a point the numerics of Section 5 confirm independently and more finely on the computed range.

Remark. Deeper boxes help at the level of the constant: optimizing exponent depths $L_p \ge 2$ within the product class (a knapsack over length increments, with per-prime factors $\lambda_{\min}\bigl(T_L\bigl(p^{-1/2}\bigr)\bigr)$) raises the constant 2 in Corollary 1 toward approximately 2.22. The shape of that optimization is derived, but the constant rests on a small-a linearization that is inexact at $p = 2, 3$; we do not claim it. It sits in any case below the ceiling of Section 3.

2.5 Machine verification

The exact content of Theorem 1 is formalized in Lean 4 against a pinned Mathlib in the companion repository [RV26], file `RiemannVenue/Kernels/PrimorialBound.lean`. The statement is deliberately eigenvalue-API-free: it is the Rayleigh-witness form `primorial_rayleigh_upper_bound`: for every finite set S of primes and every N with $\prod_{p \in S} p \le N$, there exists $x : \mathrm{Fin}\, N \to \mathbb{R}$, $x \ne 0$, with $x \cdot (K_N x) \le \left(\prod_{p \in S}(1 - p^{-1/2})\right) \cdot (x \cdot x)$,

with the two dot products evaluated exactly ($x \cdot x = 2^{|S|}$ and $x \cdot K_N x = \prod_p (2 - 2p^{-1/2})$), so the inequality is an equality read as $\le$. The supporting declarations are `gcdKernelEntry_prod_prod` (Lemma 1), `signed_powerset_double_sum` (Lemma 2), and `mobiusPrimorialProfile` (the witness); the specialization to initial segments of primes is `primesLE_rayleigh_upper_bound`. The axiom audit of the main theorem reports exactly the three standard Lean axioms `propext`, `Classical.choice`, `Quot.sound`; no extra axioms, no `sorry`. The asymptotic evaluation of Corollary 1 (Mertens, PNT, Rosser–Schoenfeld) is **not** formalized; the machine-checked claim is the exact finite inequality.

3 The product-family barrier

The primorial witness is the simplest member of a natural class: **product (tensor) test vectors**. Fix a finite prime set P and exponent lengths $L_p \ge 2$, and suppose the divisor box

$$\mathcal{B} = \left\{ n = \prod_{p \in P} p^{k_p} : 0 \le k_p \le L_p - 1 \right\}$$

fits inside $\{1, ..., N\}$, i.e. $\prod_{p \in P} p^{L_p - 1} \le N$. A product vector is $x_n = \prod_{p \in P} v^{(p)}_{v_p(n)}$ on $\mathcal{B}$ (and 0 elsewhere), with arbitrary nonzero per-prime profiles $v^{(p)} \in \mathbb{R}^{L_p}$; Liouville or Möbius signs are themselves of this form. On the box, the kernel entries factorize per prime as $K(m, n) = \prod_{p \in P} a_p^{|v_p(m) - v_p(n)|}$ with $a_p = p^{-1/2}$ (the same computation as Lemma 1 with exponents), so the Rayleigh quotient tensorizes exactly:

$$R(x) = \prod_{p \in P} \frac{v^{(p)\top}\, T_{L_p}\left(a_p\right) v^{(p)}}{\left\| v^{(p)} \right\|^2}, \quad T_L(a) = \left(a^{|i-j|}\right)_{0 \le i,j < L},$$

the Kac–Murdock–Szegő matrix. [Derived; verified numerically at machine precision on every grid point tested, see Section 5.] The following floor caps the whole class.

Lemma 3 (per-prime floor; derived). *For all* $0 < a < 1$ *and all* $L \ge 1$,

$$\lambda_{\min}(T_L(a)) \ge \frac{1 - a}{1 + a}.$$

Proof. The KMS matrix has the classical tridiagonal (AR(1) precision-matrix) inverse

$$T_L(a)^{-1} = \frac{1}{1 - a^2}\left[(1 + a^2)I - a(S + S^\top) - a^2(E_{11} + E_{LL})\right],$$

where S is the shift and E_{11}, E_{LL} are corner units, verified by direct multiplication. Gershgorin applied to $T_L(a)^{-1}$: an interior row has diagonal $(1 + a^2)/(1 - a^2)$ and off-diagonal radius $2a/(1 - a^2)$, giving $\lambda_{\max}(T_L(a)^{-1}) \le (1 + a^2 + 2a)/(1 - a^2) = (1 + a)/(1 - a)$; a border row gives the smaller value $1/(1 - a)$. Inverting, $\lambda_{\min}(T_L(a)) \ge (1 - a)/(1 + a)$.□

The floor is sharp as $L \to \infty$: the symbol of the infinite Toeplitz operator $\left(a^{|i-j|}\right)$ is the Poisson kernel $P_a(\theta) = (1 - a^2)/(1 - 2a\cos\theta + a^2)$, whose minimum is $P_a(\pi) = (1 - a)/(1 + a)$. This polydisc reading returns in Section 6.

Proposition 1 (product-family barrier; derived, not machine-checked). *Every product test vector x (any prime set P, any lengths $L_p \geq 2$, any per-prime profiles) with box inside $\{1, ..., N\}$ satisfies*

$$R(x) \geq \prod_{p \in P} \frac{1 - p^{-1/2}}{1 + p^{-1/2}}, \quad \text{hence} \quad -\log R(x) \leq (4 + o(1)) \frac{\sqrt{\log N}}{\log \log N}.$$

Proof. The first inequality is the tensorization together with Lemma 3. For the second, note $-\log[(1-a)/(1+a)] = 2 \operatorname{artanh}(a)$, so

$$-\log R(x) \leq \sum_{p \in P} 2 \operatorname{artanh}\left(p^{-1/2}\right) =: \sum_{p \in P} f(p).$$

The box constraint forces the budget $\sum_{p \in P} \left(L_p - 1\right) \log p \leq \log N$, hence (as $L_p \geq 2$) $\sum_{p \in P} \log p \leq \log N$. The function $f(p) / \log p$ is decreasing in p, so by a standard exchange argument the budgeted sum $\sum_{p \in P} f(p)$ is maximized, among all admissible P, by an initial segment: if y is minimal with $\theta(y) \geq \log N$, then $\sum_{p \in P} f(p) \leq \sum_{p \leq y} f(p)$. Since $\operatorname{artanh}(t) = t + O(t^3)$ and $\sum_p p^{-3/2}$ converges,

$$\sum_{p \leq y} f(p) = 2 \sum_{p \leq y} p^{-1/2} + O(1) = (4 + o(1)) \frac{\sqrt{y}}{\log y}$$

by the same partial summation as in Corollary 1, and $y = (1 + o(1)) \log N$ by the PNT. Substituting gives the claim. □

Thus the entire product class lives between the achieved $(2 + o(1)) \sqrt{\log N} / \log \log N$ of Corollary 1 and the ceiling $(4 + o(1)) \sqrt{\log N} / \log \log N$: **no product-type family reaches $\exp\left(-c \sqrt{\log N}\right)$ for any $c > 0$.** Two caveats for honesty. First, "product type" here means tensor profiles on full divisor boxes; tensor profiles conditioned on the simplex $\{n \leq N\}$ trade the length budget for a mean budget plus a large-deviation cost, and a first-order analysis suggests the same ceiling, but we have not pushed that to a proof. Second, Proposition 1 is a paper result; unlike Theorem 1 it has not been formalized.

The implication runs in both directions. Downward: the barrier says the theorem of Section 2 is within a factor $2 + o(1)$ (in the exponent) of everything its method can give. Upward: the numerics of the next section show the **measured** $\lambda_{\min}$ decaying strictly faster than the barrier shape on the computed window, so the true minimizer is not of product type. It must correlate the prime directions, the structural mirror of the resonance sets with which Bondarenko and Seip [BS17] beat plain Gál sets at the top edge. What replaces product structure at the bottom edge is, at present, unknown.

4 The lower bound

Everything in this section carries the label **derived**: complete paper proofs with classical inputs as marked, none of it formalized. The route is the second of the two the object offers: $1/\lambda_{\min}(K_N) = \lambda_{\max}(K_N^{-1})$, and the inverse is explicit. The derivations, with numerical spot-checks at machine precision, are recorded in the research notes `notes/lambda-min-lower-attack.md` and `notes/perron-vector-attack.md` of [RV26].

4.1 The explicit inverse and the Liouville gauge

From the Gram factorization $K_N = CC^\top$ with $C = DB\,\mathrm{diag}(\sqrt{\varphi})$ and $B^{-1}[m,d] = \mu(m/d)$ (the same object Section 5 uses computationally),

$$(K_N^{-1})_{mn} = \sqrt{mn} \sum_{k \le N,\, m|k,\, n|k} \frac{\mu(k/m)\,\mu(k/n)}{\varphi(k)}.$$

Lemma 4 (sign purification). *Fix $m, n \le N$, put $g = \gcd(m,n)$ and $l = mn/g$. Then*

$$(K_N^{-1})_{mn} = \mu(m/g)\,\mu(n/g) \cdot \sqrt{mn} \sum_{t \le N/l,\, \mu^2(t)=1,\, \gcd(t,\, l/g)=1} \frac{1}{\varphi(lt)},$$

nonzero exactly when $l \le N$ and m/g, n/g are both squarefree. Every term of the k-sum above carries the same sign, and on the support this sign is $\mu(m/g)\mu(n/g) = \lambda(m)\lambda(n)$.

Proof. Every k in the sum is a common multiple of m and n, so $k = lt$, and $k/m = (n/g)t$, $k/n = (m/g)t$. A nonzero term forces m/g, n/g and t squarefree with $\gcd(t, (m/g)(n/g)) = 1$ (otherwise one of the two ratios contains a square), and then

$$\mu(k/m)\,\mu(k/n) = \mu(n/g)\mu(t) \cdot \mu(m/g)\mu(t) = \mu(m/g)\,\mu(n/g),$$

independent of k; note that t may share primes with g, since only coprimality to $l/g = (m/g)(n/g)$ is forced. On the support m/g and n/g are squarefree, so $\mu(m/g)\mu(n/g) = (-1)^{\Omega(m)+\Omega(n)-2\Omega(g)} = \lambda(m)\lambda(n)$. □

Theorem 2 (Liouville gauge; derived). *Let $\Delta_\lambda = \mathrm{diag}(\lambda(1), ..., \lambda(N))$ and $M_N = |K_N^{-1}|$ entrywise. Then*

$$K_N^{-1} = \Delta_\lambda\, M_N\, \Delta_\lambda,$$

M_N is nonnegative and irreducible, and consequently: $\lambda_{\max}(K_N^{-1}) = \lambda_{\max}(M_N)$ exactly; $\lambda_{\min}(K_N)$ is a simple eigenvalue for every N; and the minimizing eigenvector has no zero component and sign pattern exactly $\mathrm{sign}(v_n) = \lambda(n)$.

Proof. The factorization restates Lemma 4 entrywise, and $\Delta_\lambda^2 = I$, so K_N^{-1} and M_N are similar. Irreducibility: for a prime $p \mid n$, the entry of M_N at $(n, n/p)$ is nonzero (Lemma 4 with $g = n/p$, $l = n \le N$, ratios p and 1, both squarefree); iterating over the prime factors connects every n to 1. By Perron–Frobenius for irreducible nonnegative matrices [classical], the top eigenvalue of M_N is simple with a strictly positive eigenvector u; then $v = \Delta_\lambda u$ is the top eigenvector of K_N^{-1}, that is, the bottom eigenvector of K_N, with $\mathrm{sign}(v_n) = \lambda(n)$. □

Remark. Two readings. First, the theorem upgrades the eigenvector portrait of Section 5 from measurement (mass-weighted sign agreement above 0.999) to an exact statement for every N. Second, it removes a worry: bounding $\lambda_{\max}(K_N^{-1})$ through $|K_N^{-1}|$ costs nothing, because there is no Möbius cancellation on the inverse side to lose. The bottom edge of K_N is a Perron–Frobenius problem for the explicit nonnegative matrix M_N.

4.2 The weighted Schur test

For a symmetric matrix A and any weight $w > 0$, $\lambda_{\max}(A) \le \max_m w_m^{-1} \sum_n |A_{mn}|\, w_n$ [classical]; for an irreducible nonnegative matrix the bound is attained at the Perron weight, so on M_N the only loss is the distance from w to the Perron vector. The choice of family matters: the unweighted row sums of M_N grow like $\sqrt{N}$ (the $\sqrt{n}$ column weights integrate to $\sqrt{N}$ regardless of the arithmetic), and the power family $w_n = n^{-\beta}$ caps at $\exp(-(\log 2 + o(1))\log N/\log\log N)$, pinned by rows with many prime factors. The weight that breaks through damps prime multiplicity directly:

$$w_n = \frac{\rho^{\Omega(n)}}{\sqrt{n}}, \quad \rho > 0.$$

Theorem 3 (lower bound, uniform weight; derived). *Unconditionally, as $N \to \infty$,*

$$\lambda_{\min}(K_N) \geq \exp\Bigl(-(2 + o(1))\sqrt{\log N}\Bigr).$$

With Robin's unconditional bound $\omega(m) \leq 1.3841 \log m/\log\log m$ [Rob83] in place of the prime number theorem, the constant $2\sqrt{1.3841} < 2.36$ is admissible for all N beyond an explicitly computable N_0.

Proof. Write $T_m(\rho)$ for the weighted row sum $w_m^{-1}\sum_n (M_N)_{mn} w_n$. By Lemma 4 the entries of M_N are termwise absolute sums over common multiples $k \leq N$, so, exchanging the order of summation over n and k and writing $k = ms$,

$$T_m(\rho) = m\,\rho^{-\Omega(m)} \sum_{s \leq N/m,\, \mu^2(s)=1} \rho^{\Omega(ms)-\omega(ms)} \frac{(1+\rho)^{\omega(ms)}}{\varphi(ms)},$$

where the inner sum over the divisors n of each k with k/n squarefree was evaluated exactly:

$$\sum_{n|k,\, \mu^2(k/n)=1} \rho^{\Omega(n)} = \prod_{p^e \| k} (\rho^e + \rho^{e-1}) = \rho^{\Omega(k)-\omega(k)}(1+\rho)^{\omega(k)}.$$

Split $s = uv$ with u composed of primes dividing m and v coprime to m; then $\Omega(ms) - \omega(ms) = \Omega(m) + \omega(u) - \omega(m)$ and $\omega(ms) = \omega(m) + \omega(v)$, and with $\varphi(muv) \geq \varphi(m)\varphi(u)\varphi(v)$ and the truncation $s \leq N/m$ dropped,

$$T_m(\rho) \leq \frac{m}{\varphi(m)}\,(1 + 1/\rho)^{\omega(m)} \prod_{p|m}\left(1 + \frac{\rho}{p-1}\right) \prod_{p \leq N}\left(1 + \frac{1+\rho}{p-1}\right).$$

Take logarithms, bound $\log(1+x) \leq x$, and insert four classical inputs: $\sum_{p\leq N} 1/(p-1) \leq \log\log N + B$ with an absolute constant B [Mertens; see Ten15, Ch. I.1]; $m/\varphi(m) \leq e^\gamma \log\log m + O(1/\log\log m)$ [RS62]; $\sum_{p|m} 1/(p-1) \leq (1+o(1))\log\log\log N$ for $m \leq N$ (at worst the reciprocals of the first $\omega(m)$ primes, which lie below $(1+o(1))\log N$); and $r_N := \max_{m\leq N}\omega(m) = (1+o(1))\log N/\log\log N$ (primorial inversion; PNT for the constant, or [Rob83] unconditionally). This gives

$$\log T_m(\rho) \leq \frac{r_N}{\rho} + (1+\rho)(\log\log N + B) + \rho\,(1+o(1))\log\log\log N + \log(e^\gamma \log\log N + O(1)),$$

and the choice $\rho = \sqrt{r_N/\log\log N} = (1+o(1))\sqrt{\log N}/\log\log N$ balances the two leading terms:

$$\log \lambda_{\max}(K_N^{-1}) \leq \max_m \log T_m(\rho) \leq 2\sqrt{r_N \log\log N}\,(1+o(1)) = (2+o(1))\sqrt{\log N}.$$

By Theorem 2 no factor was lost in passing to M_N. The explicit variant substitutes $r_N \le 1.3841 \log N/\log\log N$ [Rob83] throughout. $\square$

Remark. The intrinsic ceiling of the route is the Perron weight, that is, $\lambda_{\max}(K_N^{-1})$ itself; the measured looseness of the uniform-ρ test grows from 1.6 to 2.4 over $10^2 \le N \le 10^6$ (spot-check tables in the research notes). Each evaluated row-sum bound is also a certificate at its own N: at $N = 10^6$ the uniform test gives $\lambda_{\min} \ge 1/654.7 = 1.53 \times 10^{-3}$ against the true 3.70×10^{-3}.

4.3 The budget-proportional schedule

The uniform weight pays r_N/ρ because it charges every prime equally while the binding rows are exactly those with many prime factors. A per-prime weight can charge each prime in proportion to its share of $\log m$; since $\sum_{p|m} \log p \le \log m \le \log N$ for every $m \le N$, the row premium is then bounded by the budget itself, no row is exceptional, and a factor $\sqrt{\log\log N}$ comes off the exponent.

Theorem 4 (lower bound, scheduled weight; derived). *Unconditionally, as $N \to \infty$,*

$$\lambda_{\min}(K_N) \ge \exp\left(-(2+o(1))\sqrt{\log \frac{N}{\log} \log N}\right).$$

The classical inputs are Mertens' theorem, Rosser–Schoenfeld bounds for $m/\varphi(m)$, and Chebyshev-grade bounds for $\sum_{p\le x} p^{-1/2}$ and $\sum_{p>x} 1/(p\log p)$; neither the prime number theorem nor Robin's bound is used.

Proof. Let w be any multiplicative weight with $w(1) = 1$ and write $W(n) = \sqrt{n}\, w(n)$. The opening of the proof of Theorem 3 carries over unchanged: exchanging summation over n and common multiples $k = ms$ and evaluating the inner divisor sum,

$$T_m(w) = \frac{m}{W(m)} \sum_{s\le N/m,\, \mu^2(s)=1} \frac{1}{\varphi(ms)} \prod_{p^e \| ms} (W(p^e) + W(p^{e-1})).$$

Split $s = uv$ with u composed of primes dividing m and v coprime to m, use $\varphi(muv) \ge \varphi(m)\varphi(u)\varphi(v)$, and drop the truncation:

$$T_m \le \frac{m}{\varphi(m)} \prod_{p^a\|m} F_p(a) \prod_{p\le N,\, p\nmid m} G_p,$$

$$F_p(a) = 1 + \frac{W(p^{a-1})}{W(p^a)} + \frac{W(p^{a+1}) + W(p^a)}{W(p^a)(p-1)}, \quad G_p = 1 + \frac{W(p)+1}{p-1}.$$

Now take W geometric in each prime, $W(p^a) = \eta_p^a$, with the schedule

$$\eta_p = \min\left(\sqrt{p-1}, \frac{A}{\log} p\right), \quad A = \sqrt{\log N \cdot \log\log N},$$

and write $V = \sqrt{\log N/\log\log N}$, so that $1/\eta_p \ge (\log p)/A$ everywhere, with equality on the large-prime branch. Since $\sqrt{p-1}\cdot \log p$ is increasing, the minimum switches once, at the crossing y_1 solving $\sqrt{y_1 - 1}\cdot \log y_1 = A$; it satisfies $\log N/\log\log N \le y_1 \le \log N$ and

$\log y_1 = (1+o(1)) \log\log N$. For geometric W the factor $F_p(a) = 1 + 1/\eta_p + \big(\eta_p + 1\big)/(p-1)$ does not depend on a, and since $(1+x+y)/(1+y) \le 1+x$,

$$\log T_m \le \log(m/\varphi(m)) + \sum_{p|m} \log\left(1 + \frac{1}{\eta_p}\right) + \sum_{p \le N} \log G_p.$$

Each piece in turn.

- *Row premium, small primes.* $\sum_{p \le y_1} \log(1 + 1/\sqrt{p-1}) \le \sum_{p \le y_1} 1/\sqrt{p-1} = O\big(\sqrt{y_1}/\log y_1\big) = O(\sqrt{\log N}/\log\log N) = O(V/\sqrt{\log\log N}) = o(V)$ [Chebyshev-grade partial summation, classical].
- *Row premium, large primes.* For every $m \le N$, squarefree or not,

$$\sum_{p|m,\, p>y_1} \log\left(1 + \frac{1}{\eta_p}\right) \le \frac{1}{A} \sum_{p|m} \log p \le \frac{\log m}{A} \le \frac{\log N}{A} = V:$$

  the cost metric is the budget metric, so the bound is uniform in m, and deep exponents cost nothing because $F_p(a)$ is independent of a.
- *Column mass.* $\log G_p \le \big(\eta_p + 1\big)/(p-1)$. The primes $p \le y_1$ contribute $O\big(\sqrt{y_1}/\log y_1\big) = o(V)$ as above; the $+1$ contributes $\log\log N + O(1)$ [Mertens] $= o(V)$; and, with $\sum_{p>x} 1/(p \log p) = (1+o(1))/\log x$ [classical, partial summation on Mertens],

$$\sum_{p>y_1} \frac{\eta_p}{p-1} = \sqrt{\log N \log\log N} \cdot \frac{1+o(1)}{\log y_1} = (1+o(1))\, V,$$

  since $\log y_1 = (1+o(1)) \log\log N$.
- *The prefactor.* $\log(m/\varphi(m)) = O(\log\log\log N)$ uniformly for $m \le N$ [RS62], again $o(V)$.

Collecting, $\log T_m \le (2+o(1))V$ uniformly in $m \le N$, so $\lambda_{\max}(K_N^{-1}) \le \exp\Big((2 + o(1))\sqrt{\log N/\log\log N}\Big)$, and by Theorem 2 nothing was lost in passing to M_N. □

Remark. Certificates improve as well: the one-parameter family $\eta_p = \min\big(\sqrt{p-1}, A/\log p\big)$ with $A = 0.7\sqrt{\log N \log\log N}$ beats the uniform family at every computed N; at $N = 10^6$ it gives $\lambda_{\min} \ge 1/554.2 = 1.80 \times 10^{-3}$. Its binding rows are primorials, and its compensated value $\log T/\sqrt{\log N}$ declines ($1.749 \to 1.700$ over $10^3 \le N \le 10^6$) where the uniform family's stayed flat near 1.75: the signature of leaving the $\exp\big(c\sqrt{\log N}\big)$ class.

Remark. On the constant, and on the class. Within the scheme of the proof the 2 is a forced balance: row premium and column mass both equal $(1+o(1))V$ at the optimal schedule, and the small-prime block forces $\log y_0 \le (1+o(1)) \log\log N$, so no geometric schedule goes below $2\sqrt{\log N/\log\log N}$. Whether the full multiplicative class can go lower is a different question, and numerically no barrier is visible: minimizing $\max_m T_m(w)$ over all multiplicative weights is a convex problem in the prime-power log-increments (a log-sum-exp of affine functions, so every evaluated weight is a rigorous certificate), and the global optimum at $N = 10^3$–10^6 lands within 7–10% of $\lambda_{\max}(K_N^{-1})$ itself, with the compensated gap flat across three decades [numerical; research notes]. The remaining distance from the theorem to the truth is the distance from geometric schedules to the Perron vector of M_N, whose per-prime profiles are factorial-damped rather than geometric and whose breadth follows the same $1/\log p$ plateau as the schedule above [numerical]. The shape and constant $\exp\Big(2\sqrt{\log N/\log\log N}\Big)$

also appear in [ABS15] as the floor for critical GCD sums over extremal sets; the extremal arithmetic on both sides is the same budget-constrained allocation of $\log N$ across the primes.

4.4 The rate

Corollary 2 (the exponent; derived). *Unconditionally,*

$$\exp\left(-(2+o(1))\sqrt{\log\frac{N}{\log}\log N}\right) \le \lambda_{\min}(K_N) \le \exp\left(-0.99\,\frac{\sqrt{\log N}}{\log}\log N\right) \quad \text{(the right side for } N \ge 3.3\times 10^7\text{)},$$

hence $-\log\lambda_{\min}(K_N) = (\log N)^{1/2+o(1)}$*, and* $\lambda_{\min}(K_N) = \exp(-(c+o(1))\sqrt{\log N})$ *holds for no constant* $c > 0$*.*

Proof. Combine Theorem 4 with Corollary 1 and its explicit variant; for the exponent, $\sqrt{\log N}/\log\log N = (\log N)^{1/2-\log\log\log N/\log\log N}$. For the exclusion, the lower bound forces $-\log\lambda_{\min}/\sqrt{\log N} \le (2+o(1))/\sqrt{\log\log N} \to 0$. $\square$

The gap is now a corridor: $0.99\sqrt{\log N}/\log\log N \le -\log\lambda_{\min} \le (2+o(1))\sqrt{\log N/\log\log N}$, of relative width $\sqrt{\log\log N}$. Where each side loses is identified: the upper bound on $\lambda_{\min}$ gives away only what lies beyond the product class (Section 3), the lower bound only the distance from multiplicative weights to the Perron vector of M_N, measured at 7–10% of the logarithm at computable N. The numerics of the next section measure the window value of the exponential fit at ≈ 1.36–1.41; the corollary makes that value a finite-size effect.

5 Numerics to $N = 1.3\times 10^7$

All computations in this section are **numerical** in the sense of the label convention: deterministic, committed artifacts (scripts `scripts/lambda_min_lanczos.py` and `scripts/lambda_min_testfamily.py`, seed 20260708, and the executed notebook `notebooks/lambda-min-rate.ipynb` in [RV26]), with no claim beyond the computed range.

5.1 The exact Möbius-sparse inverse

Dense symmetric eigensolvers reach $N \approx 2.5\times 10^4$ on this problem (the last dense anchor below took a single ~ 5 GB solve). The route past that ceiling is the arithmetic of the kernel itself. The Gram factor $C = DB\,\mathrm{diag}(\sqrt{\varphi})$ of Section 2 is unit-triangular up to diagonal scaling, and the inverse of the divisibility matrix B is the Möbius matrix **on the same sparsity pattern**:

$$B^{-1}[m,d] = \mu(m/d) \text{ for } d \mid m, \quad \text{hence} \quad C^{-1} = \mathrm{diag}\big(\varphi^{-1/2}\big)\,B^{-1}\,\mathrm{diag}\big(\sqrt{n}\big),$$

explicitly available with approximately $(6/\pi^2)N\log N$ nonzeros. Then $K_N^{-1}x = C^{-\top}(C^{-1}x)$ costs two sparse matrix–vector products, and Lanczos iteration (ARPACK `eigsh`, `which='LA'`, $k=2$) on K_N^{-1} targets $1/\lambda_{\min}$ as a **dominant** eigenvalue: no shift-invert, no dense algebra anywhere. The top edge $\lambda_{\max}$ comes from plain Lanczos on K_N. At $N = 13\,107\,200$ the factor has 2.2×10^8 nonzeros and the bottom-edge solve takes minutes on a laptop-class machine.

Validation, before trusting any new point:

- against dense `eigvalsh` anchors: relative differences 4.9×10^{-8} at $N = 800$, 9.1×10^{-6} at 3200, 1.1×10^{-5} at 12 800, 2.5×10^{-7} at 25 600 (agreement to the full quoted precision of the anchors);

- the algebraic identity $C^{-1}(Cx) = x$ holds to relative error $\leq 2.6 \times 10^{-16}$ on random vectors at every N;
- eigen-residuals are measured on K_N itself, not on the inverse: $\|Kv - \lambda_{\min} v\| \leq 2.7 \times 10^{-14}$ across the grid (top edge $\leq 2.1 \times 10^{-12}$), certifying each reported $\lambda_{\min}$ independently of Lanczos convergence heuristics.

One caveat is worth stating: the bottom edge is an accumulation edge and the cluster tightens slowly ($\lambda_2/\lambda_{\min}$ falls from 1.52 at $N = 800$ to 1.31 at $N = 1.3 \times 10^7$); the $k = 2$ computation returns both edge eigenvalues, and the residuals above are what certify the values.

5.2 The portrait that suggested the theorem

The dense window already contains the paper's mechanism. At $N = 2000$ the minimizing eigenvector concentrates on small, smooth, highly divisible integers: the largest components sit at $n = 6, 30, 2, 12, 10, 42, 60, 3, 4, 18, \ldots$, with 78% of the mass on $n \leq 100$ and only 1% on the entire upper half. Its **signs follow the Liouville function** $\lambda(n) = (-1)^{\Omega(n)}$ with mass-weighted agreement above 0.999 (Figure 1, right panel), the pattern Theorem 2 proves exact for every N. The minimizer is a Liouville-signed measure on smooth numbers: a finite-rank shadow of a $1/\zeta$-type mollifier (recall $\sum \lambda(n) n^{-s} = \zeta(2s)/\zeta(s)$), not a truncation-boundary artifact. A useful classical benchmark calibrates the scale: pinning $x_1 = 1$ gives the exact identity

$$\min_{x:\, x_1 = 1} x^\top K_N x = \frac{1}{(K_N^{-1})_{11}} = \frac{1}{\sum_{k \leq N} \mu^2(k)/\varphi(k)} = \frac{1}{\log N + 1.3326\ldots + o(1)},$$

the Selberg-sieve sum [classical; verified numerically to 10^{-6}]. The measured $\lambda_{\min}$ sits roughly one full extra logarithm below this pinned value on the dense window: the free minimization buys more than the sieve's single log, which is what Theorem 1 explains and exceeds.

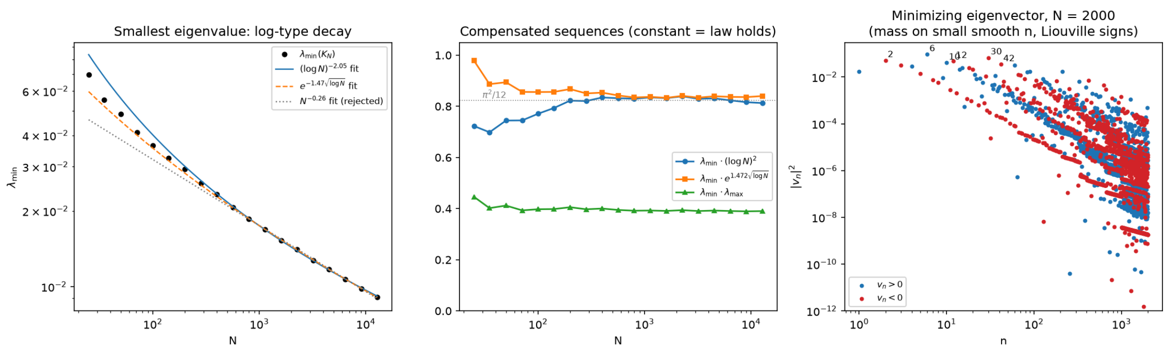

Figure 1: The dense-spectrum window $N \leq 25\,600$. Left: $\lambda_{\min}(N)$ with fitted laws; the fixed power of N is rejected. Middle: compensated sequences for the two surviving two-parameter laws. Right: the minimizing eigenvector at $N = 2000$, with mass on small smooth n and signs following the Liouville function.

5.3 Results and law discrimination

Selected values on the half-octave grid $800 \cdot 2^{k/2}$ (28 points, $800 \leq N \leq 13\,107\,200$) appear in Table 1; the dense anchors below $N = 25\,600$ ($\lambda_{\min} = 0.018566$ at 800, 0.012735 at 3200, 0.009088 at 12 800, 0.0077544 at 25 600) join the grid smoothly.

N	$\lambda_{\min}$	$\lambda_{\min}\cdot(\log N)^2$	$\lambda_{\min}\cdot e^{1.472\sqrt{\log N}}$	$\lambda_{\min}\cdot\lambda_{\max}$
51 200	0.0066590	0.7830	0.8483	0.3893
102 400	0.0057524	0.7656	0.8535	0.3889
204 800	0.0049992	0.7477	0.8601	0.3888
409 600	0.0043680	0.7295	0.8678	0.3889
819 200	0.0038332	0.7107	0.8760	0.3890
1 638 400	0.0033769	0.6914	0.8846	0.3891
3 276 800	0.0029862	0.6721	0.8938	0.3892
6 553 600	0.0026502	0.6529	0.9036	0.3893
13 107 200	0.0023598	0.6338	0.9138	0.3895

Table 1: Bottom-edge values from Lanczos on the exact sparse inverse (selected rows; eigen-residuals $\le 2.7\times10^{-14}$ on K_N). The constant 1.472 in the fourth column is the dense-window fit, kept fixed to expose the drift.

The law $c(\log N)^{-2}$ is rejected on the computed range. On the dense window this law fits well: $\lambda_{\min}\cdot(\log N)^2$ is constant to $\pm3\%$ up to $N=25\,600$, and the sieve heuristics of the pinned identity make it structurally attractive. The extension kills it: the compensated sequence falls monotonically from its plateau ≈ 0.83 to 0.634 at $N=1.3\times10^7$ (a 24% decline with no flattening), and a $(\log N)^{-2}$ extrapolation calibrated at the $N=25\,600$ anchor overshoots the measured value at $N=13\,107\,200$ by +26.1%. The local log-power exponent, which the law requires to settle at 2, instead rises monotonically: 2.1–2.2 on the dense window, 2.31 near $N\approx4\times10^4$, 2.69 on the top octave; free-exponent fits find no stable value ($2.31\to2.48\to2.59$ as the window moves up, with rms residuals three times worse than the exponential law in every window).

The law $A\exp\left(-c\sqrt{\log N}\right)$ fits each window, with a slowly drifting c. Refitting by window on the measured grid: $c=1.411$ (all $N\ge800$), $c=1.379$ ($N\ge25\,600$), $c=1.358$ ($N\ge409\,600$, rms 0.0008 in $\log\lambda$); the local rate declines by roughly 0.010 per octave through 1.341 at the top octave. One internal consistency check: this law predicts a local log-power exponent $(c/2)\sqrt{\log N}$, which at the top octave gives 2.686, against 2.686 measured. The log-power drift is exactly what the exponential law forces. The decline of c is the finite-size effect Corollary 2 forces.

The product-family (barrier-shaped) law loses decisively. Proposition 1 makes $A\exp(-C\sqrt{\log N}/\log\log N)$ the natural law for product families. Fitted on the measured $\lambda_{\min}$ (13-point grid; rms in $\log\lambda$):

window	$Ae^{-c\sqrt{\log N}}$	$Ae^{-C\sqrt{\log N}/\log\log N}$	$A(\log N)^{-a}$
$N\ge800$ (all 13)	$c=1.406$ (**0.0097**)	$C=20.2$ (0.164)	$a=2.32$ (0.031)
$N\ge25\,600$	$c=1.378$ (**0.0036**)	$C=16.1$ (0.030)	$a=2.48$ (0.0092)
$N\ge409\,600$	$c=1.358$ (**0.0009**)	$C=14.4$ (0.0062)	$a=2.59$ (0.0027)

Table 2: Two-parameter law fits to the measured $\lambda_{\min}$ (rms residual in $\log\lambda$ in parentheses). The barrier-shaped law is 7–17 × worse than the exponential law in every window and needs prefactors $A\sim10^6$–10^{10}; the function $\sqrt{\log N}/\log\log N$ is nearly flat across the whole window while $\lambda_{\min}$ falls by a factor 7.9.

Three-parameter fits refine rather than overturn this picture: the scan $\lambda \approx A \exp\big(-c(\log N)^{\theta}\big)$ prefers an effective exponent slightly **below** $1/2$, $\theta \approx 0.36$–0.38 (the direction Theorem 4 forces), and the hybrid $A(\log N)^{-\beta} e^{-c\sqrt{\log N}}$ gives $\beta \approx 0.6$–0.7, $c \approx 1.0$. On any feasible window these cannot be separated from slowly varying corrections inside the square root (the ABS-type $\log\log$ factors); discriminating them would need local-rate resolution of $\sim 0.5\%$ per octave at $N \gtrsim 10^9$. What the window **does** decide is the two rejections above.

Finally, the product families of Section 3 were implemented exactly (three families; per-prime product formula, direct double sum, sieve identity, and a full sparse matvec agreeing to 4.2×10^{-15}): the best product-family value tracks the true $\lambda_{\min}$ within a small factor that grows slowly and monotonically: 1.72 at $N = 800$, 2.07 at 12 800, 2.51 at 102 400, 3.09 at 1.6×10^6, 3.50 at 1.3×10^7, an average drift of about $+5.2\%$ per octave. The true minimizer strictly outruns the entire product class at a slow, polylog-compatible pace: the measured gap neither flattens (which would certify the box shape) nor accelerates past every polylog (which Corollary 2 forbids in the limit).

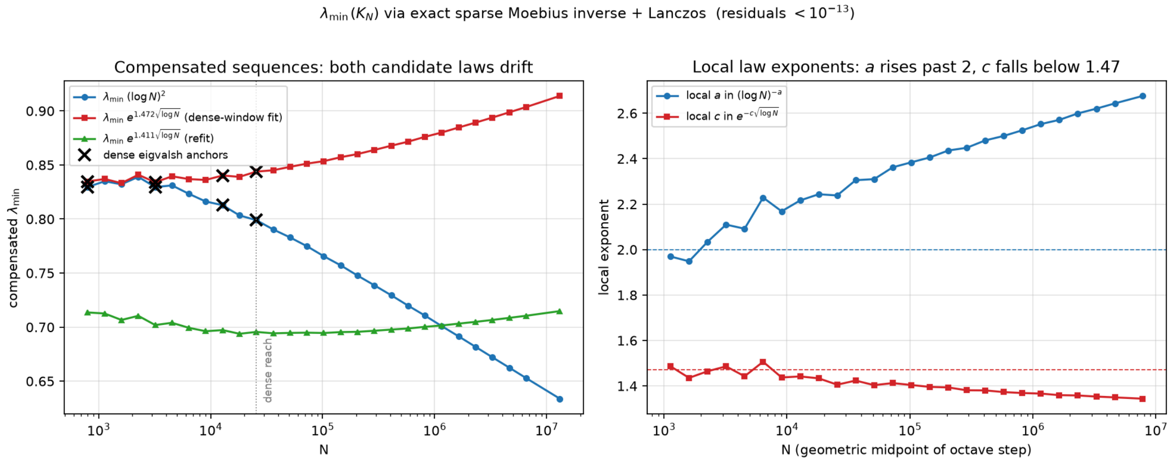

Figure 2: The Lanczos extension, $800 \le N \le 1.3 \times 10^7$, via the exact Möbius-sparse inverse. Left: both two-parameter candidate laws drift; $\lambda_{\min}(\log N)^2$ falls with no flattening while the dense-window-calibrated exponential compensation rises slowly. Crosses mark dense `eigvalsh` anchors. Right: the local log-power exponent rises past 2 as the exponential law forces, while the local exponential rate drifts slowly below 1.47.

6 The remaining gap, and the reciprocal edges

6.1 The corridor

Corollary 2 settles the exponent and rules out one law; what it leaves open is the slowly varying factor. The walls stand at

$$0.99\,\frac{\sqrt{\log N}}{\log\log N} \le -\log \lambda_{\min}(K_N) \le (2 + o(1))\sqrt{\log \frac{N}{\log}\log N},$$

a corridor of relative width $\sqrt{\log\log N}$, and the question is where inside it the truth lives:

Does $-\log \lambda_{\min}(K_N)/\sqrt{\log N/\log\log N}$ tend to a constant, sink toward the lower wall, or drift between the two?

The pure law $\exp(-(c+o(1))\sqrt{\log N})$ is no longer among the options. On the computed window the exponential-fit constant sits at $c \approx 1.36$–1.47, and the local rate's measured decline (≈ 0.010 per octave) is the visible part of a drift Corollary 2 leaves no way around. In the corridor's own units the measured value $-\log \lambda_{\min}/\sqrt{\log N/\log\log N}$ runs from 2.13 at $N = 800$ to 2.50 at $N = 1.3 \times 10^7$, rising, with no tension against either wall. We still decline to conjecture the finer structure: window-effective constants should not be over-trusted, since $\log\log$-type factors inside the exponent are unresolvable at any feasible N (Section 5). What would move each side is identified in Section 3 and Section 4: improving the upper bound on $\lambda_{\min}$ means leaving the product class, which the barrier shows is exhausted; improving the lower bound means closing the distance from multiplicative weights to the Perron vector of $M_N = |K_N^{-1}|$, whose top eigenvalue is the answer itself (Theorem 2) and which the full multiplicative class already matches to 7–10% at computable N. Both sides are now questions about the same object, the Perron vector of one explicit nonnegative arithmetic matrix. The Poisson-integral representation of [ABS15], which bounds the quadratic form from below by integrals that sieve methods can evaluate, remains the one known route not yet exploited on the lower side.

6.2 The Liouville conjugation symmetry

On smooth supports the kernel is a polydisc (prime-torus) object: $K(m,n) = \prod_p a_p^{|v_p(m)-v_p(n)|}$ with $a_p = p^{-1/2}$, and per prime the Toeplitz symbol of $\left(a^{|i-j|}\right)$ is the Poisson kernel $P_a(\theta)$, with maximum $P_a(0) = (1+a)/(1-a)$ at the point $\theta = 0$ and minimum $P_a(\pi) = (1-a)/(1+a)$ at the half-turn $\theta = \pi$, and the **exact reciprocity** $P_a(0)\cdot P_a(\pi) = 1$. The half-turn of every prime circle is implemented on the matrix by an exact symmetry:

Proposition 2 (Liouville conjugation; machine-checked). *Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function and $L_N = \mathrm{diag}(\lambda(1), \ldots, \lambda(N))$. Then: (i) $(L_N K_N L_N)(m,n) = \lambda(m)\lambda(n)K_N(m,n)$ entrywise; (ii) $L_N^2 = I$, so $K_N \mapsto L_N K_N L_N$ is an involutive similarity (in particular the conjugated kernel has the same spectrum as K_N and is again positive semidefinite); (iii) on smooth supports the conjugated kernel is $\prod_p \left(-a_p\right)^{|v_p(m)-v_p(n)|}$: the substitution $\theta_p \mapsto \theta_p + \pi$ in every prime coordinate, under which $P_a(\theta+\pi) = P_{-a}(\theta)$.*

Proof. (i) and (ii) are immediate from $\lambda(n)^2 = 1$. For (iii), the point is a parity identity: pointwise $|a-b| \equiv a+b \pmod 2$, so over any prime set S covering the supports of m and n,

$$\prod_{p\in S}(-1)^{|v_p(m)-v_p(n)|} = (-1)^{\sum_p v_p(m)}(-1)^{\sum_p v_p(n)} = \lambda(m)\,\lambda(n),$$

since $\sum_p v_p(m) = \Omega(m)$. Multiplying the unsigned factorization $\prod_p a_p^{|v_p(m)-v_p(n)|} = \gcd(m,n)/\sqrt{mn}$ by this sign gives $\prod_p \left(-a_p\right)^{|\cdot|} = \lambda(m)\lambda(n)\gcd(m,n)/\sqrt{mn}$, which is (i) restricted to the smooth support. $\square$

The signed-kernel identity of the proof is machine-checked as `prod_neg_rpow_half_factorization_eq_liouville_mul_gcd_div_sqrt` in `RiemannVenue/Kernels/LiouvilleConjugation.lean` [RV26], with the parity lemma (`prod_neg_one_pow_natAbs_factorization`), the matrix-level statements (`liouvilleDiagonal_mul_normalizedGcdKernel_mul_apply`, `liouvilleDiagonal_mul_self`), and positivity of the conjugated kernel (`liouvilleDiagonal_conj_posSemidef`).

The relevance to the two spectral edges: the top eigenvector of K_N is Perron-positive (all $\theta_p = 0$), while the bottom eigenvector carries exactly the Liouville signs (all $\theta_p = \pi$), first measured (mass-weighted sign agreement 1.000000 at $N = 25\,600$), then proved for every N (Theorem 2). The conjugation Proposition 2 exchanges the two faces, and per prime the edge values obey the exact reciprocity $P_a(0)P_a(\pi) = 1$.

6.3 The constant product of the edges

Observation 1 (numerical only). *On the computed grid,* $\lambda_{\min}(N) \cdot \lambda_{\max}(N) \in [0.3887, 0.3895]$ *for all* $2.05 \times 10^5 \le N \le 1.31 \times 10^7$*, constant to* $\pm 0.1\%$ *(with a very slight upward drift, about* $+0.02\%$ *per octave at the top), while each factor separately moves by a factor of about* 1.9*. Over the full range* $283 \le N \le 1.3 \times 10^7$ *the product stays within* $\pm 1\%$ *of* 0.389*.*

We record this as the most robust unexplained regularity in the data set, and we explicitly do not claim a limit theorem: the observation is numerical, full stop. The natural reading runs through Proposition 2: if both extremal vectors were exactly of product type with deep per-prime boxes, the product of the per-prime edge pairs would tend to $\prod_p P_{a_p}(0)P_{a_p}(\pi) = 1$ over the active primes, and the measured constant would be the **truncation deficit** of this reciprocity: the finite-box, finite-N shortfall from the exact polydisc identity. The data support the mechanism at the level of sign patterns (exact Liouville signs at the bottom, Perron positivity at the top), shared smooth support (the top-ten supports of the two edge vectors share six of ten elements at $N = 25\,600$), and the flatness of the product; they also bound its reach: the two magnitude profiles are not mirror images (cosine similarity 0.68; the top vector spreads further into deeper prime powers), and the product-family analogue of the constant comes out flat but wrong (0.21–0.25). Consistently with Section 3: boxes are not the true extremal vectors, and whatever correlates the primes at the bottom edge apparently does so at the top edge in a reciprocal way. Explaining the constancy of $\lambda_{\min} \cdot \lambda_{\max}$, even heuristically but quantitatively, is posed here as an open problem alongside the corridor problem above; by Theorem 2 the two are questions about the same Perron vector.

7 Related work

GCD sums and the top of the spectrum. Gál's theorem [Gal49] settles $\alpha = 1$ with the optimal $(\log\log N)^2$; the critical case $\alpha = 1/2$ runs through Dyer–Harman [DH86] to Aistleitner–Berkes–Seip [ABS15], who solved the Dyer–Harman problem via a Poisson-integral representation on the polydisc and recorded the lower obstruction $\exp\left(2\sqrt{\log N/\log\log N}\right)$, and to Bondarenko–Seip [BS15, BS17], whose resonance-set refinements pinned the optimal exponent shape $\sqrt{\log N \log\log\log N/\log\log N}$ with consequences for extreme values of $|\zeta(1/2 + it)|$. These results govern extremal **sets**; on the interval, the top edge is far smaller (Hilberdink's multiplicative-Toeplitz program [Hil17] gives the $\alpha = 1$ interval asymptotics; on the computed window at $\alpha = 1/2$ we observe only polylogarithmic growth, drifting upward; see Section 5). The bottom of the spectrum is not addressed in any of these works.

Positive definiteness and the compact regime. Lindqvist–Seip [LS98] proved positive definiteness with sharp zeta-function eigenvalue bounds for $\alpha > 1/2$, stopping exactly at the critical exponent, where $\sum_p p^{-2\alpha}$ diverges (we cite [LS98] partly second-hand, through [ABS15] and the survey literature, as the original was not accessible to us in full text).

Hilberdink–Pushnitski [HP22] (and the sequel arXiv:2401.06892, with Beurling-prime connections) give complete spectral asymptotics for the two-parameter family $E(\sigma, \tau)$ strictly inside the compact regime $\rho = \tau - 2\sigma > 0$; our kernel is exactly the excluded boundary case $E(1/2, 1)$.

Smallest eigenvalues of GCD-type matrices. Hong and Loewy [HL04] began the asymptotic study of smallest eigenvalues for unnormalized power-GCD matrices $((i,j)^{\varepsilon})$, including a conjecture (later proved) on when $\lambda_{\min}$ stays bounded away from zero; explicit bounds for the extreme eigenvalues of GCD and LCM matrices on $\{1, ..., n\}$ in terms of arithmetic functions are available (e.g. arXiv:1408.3113). None of these concern the critical normalization, where $\lambda_{\min} \to 0$ and the rate is the question.

Position of this paper. To the author's knowledge, the decay rate of $\lambda_{\min}(K_N)$ for the interval at the critical exponent had not been posed or bounded in the literature; this paper poses it, bounds it on both sides, and settles the exponent. The construction of Section 2 is best described as implicit-in-Gál on the dual side: the support is Gál's classical extremal set; the Möbius signing and the bottom-edge application are new. The gauge identity of Section 4 appears to have no antecedent either. The claim of novelty is a literature search, not a theorem, and is offered with that hedge.

8 Acknowledgments and artifacts

This work was carried out in collaboration with two AI systems, credited here with their roles: **Claude Fable 5** (Anthropic) co-developed the Lean formalizations, the computational experiments, and the research notes in interactive sessions, under the author's direction and scope decisions throughout; **GPT 5.5 Pro** (OpenAI) provided feedback on the accompanying essay and on the repository's initial outline. Correctness of the machine-checked claims rests on the Lean kernel and on reproducibility, not on trust in any author, human or artificial.

All machine-checked claims in this paper cite exact declaration names in the companion repository [RV26] (`https://github.com/idolum-ai/riemann-venue`): Lean 4 against a pinned Mathlib, zero-`sorry` continuous integration, axiom audits on the named theorems (`propext`, `Classical.choice`, `Quot.sound`). The numerical results are reproducible from deterministic committed scripts (`scripts/lambda_min_lanczos.py`, `scripts/lambda_min_testfamily.py`, seed 20260708) and an executed notebook (`notebooks/lambda-min-rate.ipynb`); the figures are committed outputs of those artifacts.

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